

# MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

**KCSE PREDICTION SERIES 1**



**232/2**

**Physics (Theory)**

**Paper 2**

**Time: 2 Hours**

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

*Kenya Certificate of Secondary Education.*

## **Instructions to candidates**

- This paper consists of two sections A and B.
- Answer **ALL** the questions in sections A and B in the spaces provided.
- **ALL** working **MUST** be clearly shown.
- Mathematical tables and silent electronic calculators may be used.
- This paper consists of 12 printed pages
- Take: Acceleration due to gravity  $g = 10 \text{ ms}^{-2}$ .
- Density of mercury  $\equiv 13600 \text{ kg/m}^3$

## **FOR EXAMINERS USE ONLY**

| SECTION | QUESTION | MAX MARKS | SCORE |
|---------|----------|-----------|-------|
| A       | 1 – 13   | 25        |       |
| B       | 14 – 17  | 55        |       |
| TOTAL   |          | 80        |       |

**SECTION A– 25 MARKS**

1. The figure 1 below shows a pinhole camera.

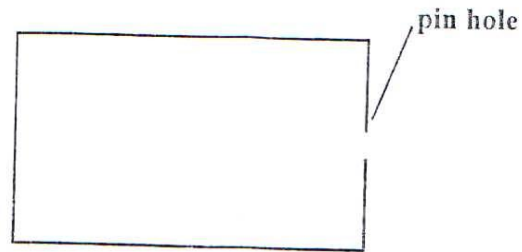


Fig 1

Sketch rays to show the information of a diminished image in the camera. Label both the object and the image.  
**(2 Marks)**

2. The figure 2 below shows the displacement-time graph for a certain wave.

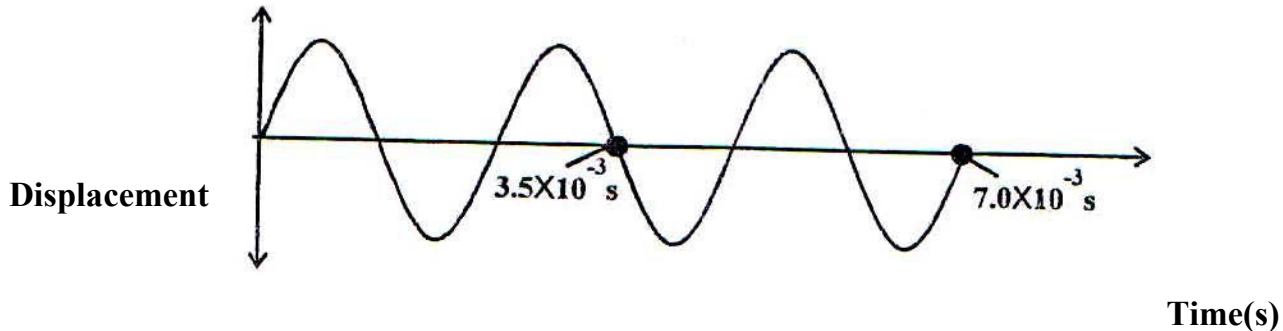


Fig 2

Determine the frequency of the wave.

**(3 Marks)**

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3. A metal of resistance  $R_1$  is rated P watts, V volts while another of resistance  $R_2$  is rated 2P watts volts  $\frac{V}{2}$  volts. Determine  $\frac{R_1}{R_2}$  (3 Marks)

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4. A leaf electroscope A is charged and placed on the bench. Another unchanged leaf electroscope B is placed on the same bench and moved close to A until the caps touch. State and explain what is observed on the leaves A and B. (2 Marks)

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5. An unmagnetised steel rod is clamped facing North-South direction and then hammered respectively for sometime. When tested, it is found to be magnetized. Explain this observation. (2 Marks)

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6. Give a reason why it is necessary to leave the caps of cells open when charging an accumulator. (1 Mark)

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7. An electromagnet is made by winding insulated copper wire on an iron core. State the two changes that could be made to increase the strength of the electromagnet. **(2 Marks)**

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8. Determine the speed of light in water given the speed of light in air is  $3.0 \times 10^8\text{m/s}$  and the refractive index of water is 1.33. **(2 Marks)**

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9. In an x-ray tube it is observed that the intensity of x-rays increases when potential differences across the filament is increased. Explain this observation. **(2 Marks)**

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10. The figure 3 below shows P-N junction connected to a battery. It is observed that the current in figure (a) is greater than in figure (b).

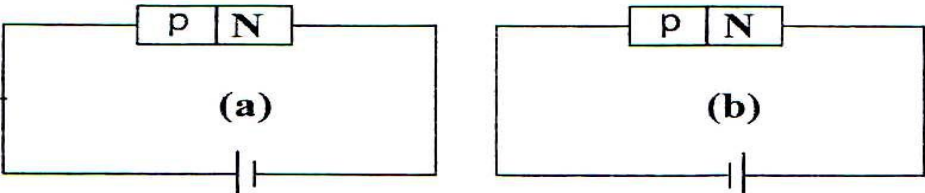


Fig 3

Explain this observation. **(2 Marks)**

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11. Explain why A.C electric power is transmitted over long distance at high voltages.(1 Mark)

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12. The wavelength of a radio wave is 1km. Determine its frequency. Given that  $V=3.0 \times 10^8$   
(2 Marks)

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13. Identify the problem in the circuit below. (1 Mark)

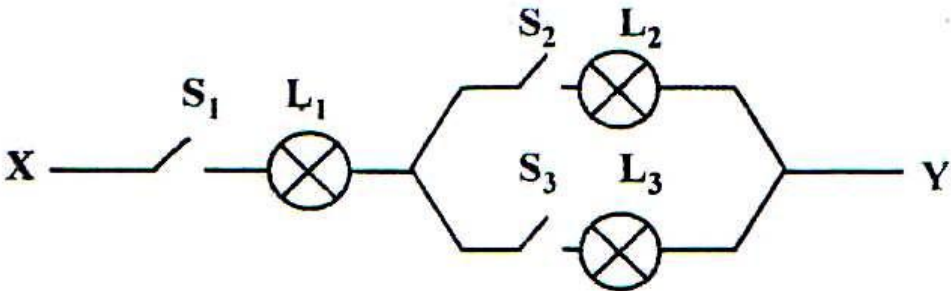


Fig 4

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**SECTION B – 55 MARKS**

14.(a) Figure below shows a pair of parallel plates of a capacitor connected to a battery. The upper plate is displaced slightly to the left.

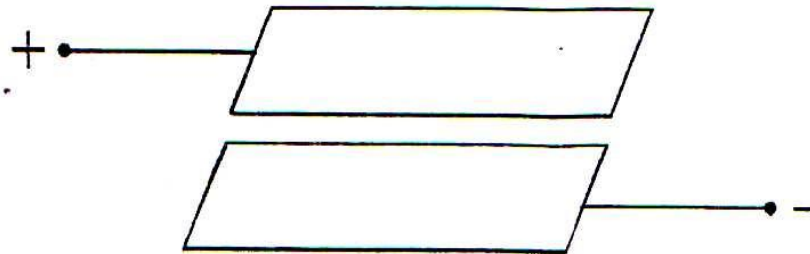


Fig 5

State with reason the effect of this movement on the capacitance. (2 Marks)

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(b) Figure 6 below is electrical circuit with three capacitors A, B and C of capacitance  $4.0\mu\text{F}$ ,  $5.0\mu\text{F}$  and  $3.0\mu\text{F}$  respectively connected to 12V battery.

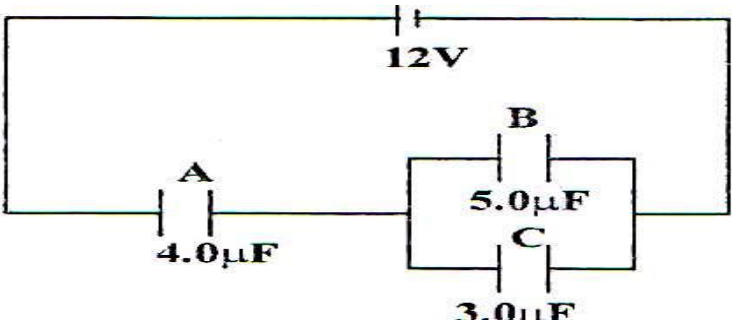


Fig 6  
Determine

(i) the combined capacitance of the three capacitors. (3 Marks)

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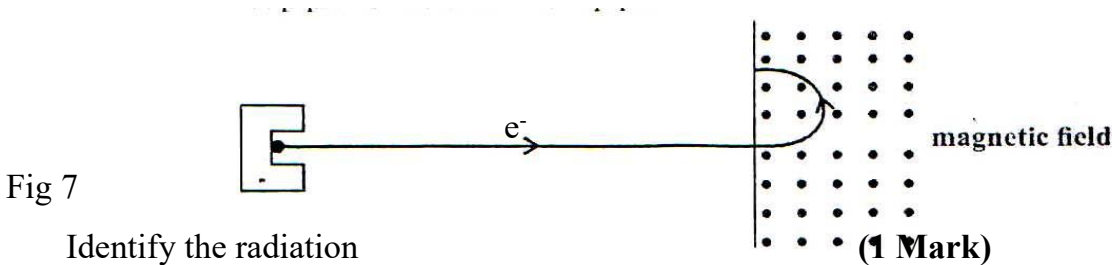
(ii) the charge stored in capacitor A. (2 Marks)

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(iii) the potential difference across the capacitor B. (2 Marks)

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15.(a) Figure below 7 shows the path of radiation from a radioactive source. The field is perpendicular to the paper and directed out of the paper.



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(b) (i) State the effect of the radiation on the gas inside the Geiger-Muller tube. (1 Mark)

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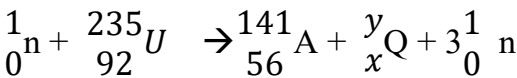
(ii) Explain how the large discharge current is created. (2 Marks)

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(c) The following is a nuclear equation for a fission process resulting from the reaction of a neutron with a Uranium nucleus.



(i) Determine the values of x and y. (2 Marks)

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(ii) State the source of the energy released. (1 Mark)

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(iii) Explain how this reaction is made continuous in a nuclear reactor. (2 Marks)

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16. (a) Figure 8 below shows ultra-violet light striking a polished zinc plate placed on a negatively charged gold-leaf electroscope.

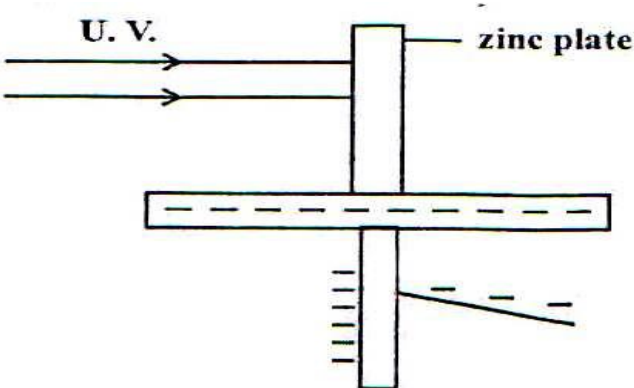


Fig 8

Explain the following observations

(i) The leaf of the electroscope falls (2 Marks)

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(ii) When the same experiment was repeated with a positively charged electroscope the leaf did not fall.

(2 Marks)

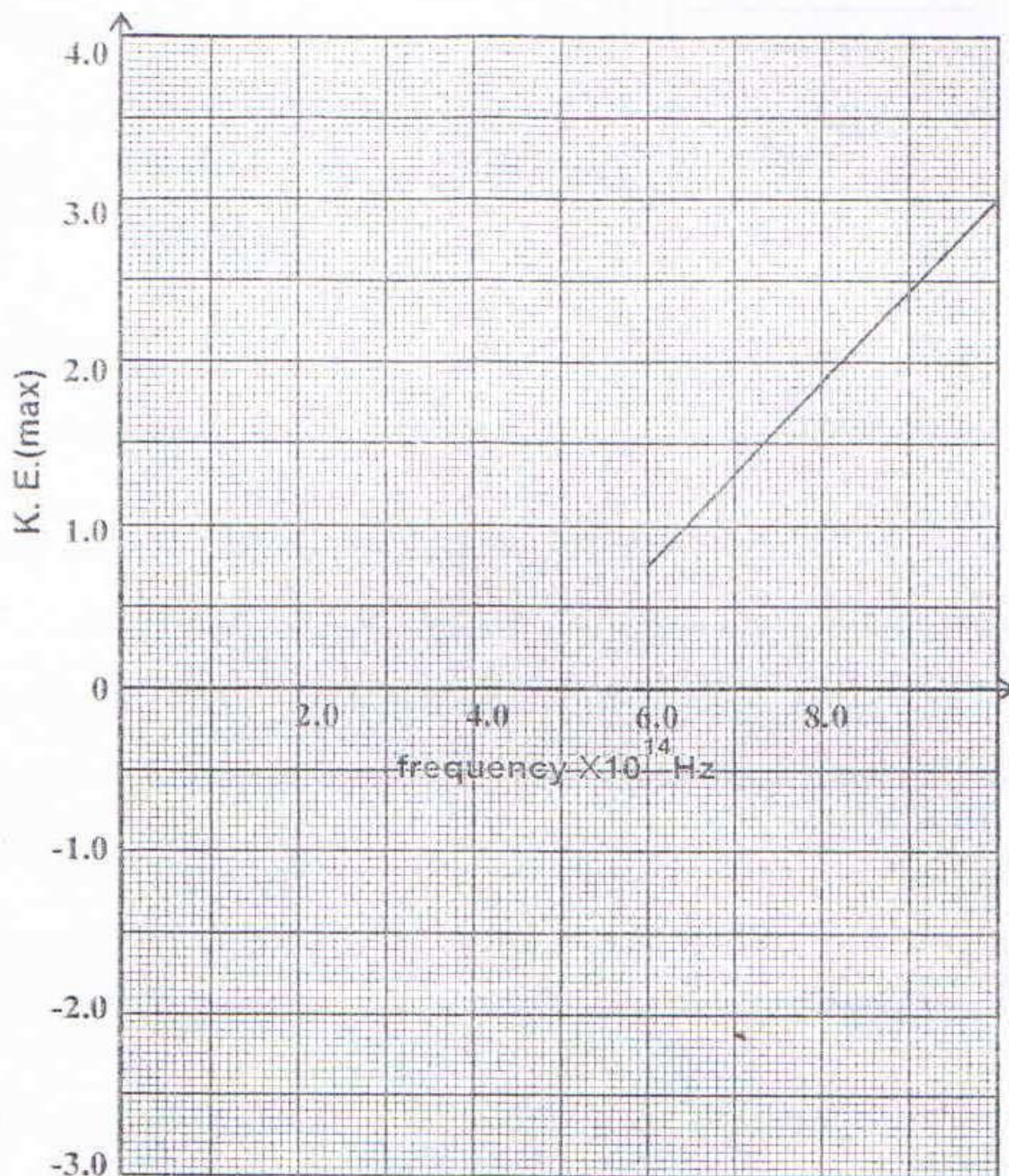
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(b)(i) State two factors which determine the speed of photoelectrons emitted by a metal surface.

(2 Marks)

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- (ii) In an experiment using a photocell, U.V light of varying frequency but constant intensity was made to strike a metal surface. The maximum kinetic energy ( $K.E_{\max}$ ) of photoelectrons for each frequency,  $f$ , measured. The graph shows how  $K.E_{\max}$  varies with  $f$ .



Given that  $K.E_{\text{max}} = hf - Q$ , determine the value of the constants  $h$  and  $Q$  from the graph.

**(6 Marks)**

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(c) Light of frequency  $5.5 \times 10^{14}\text{HZ}$  is made to strike a surface whose work function is  $2.5\text{ev}$ . Show that photoelectric effect will not take place (Use the value of  $h$  from (b) (ii) above) **(4 Marks)**

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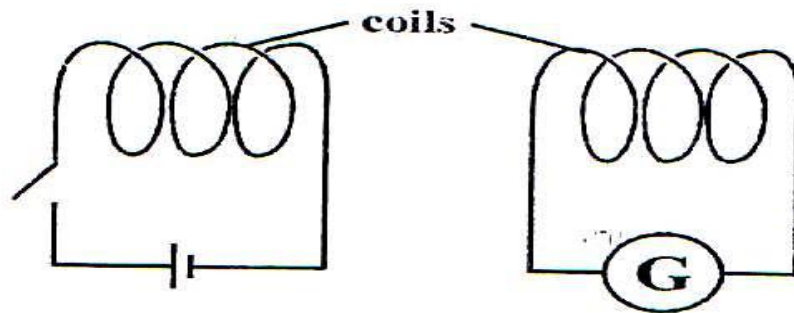
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17.(a) Figure 9 below shows two circuits close to each other.



**Fig 9**

When the switch is closed, the galvanometer shows a reading and then returns to zero. When the switch is then opened, the galvanometer shows a reading in the opposite direction and then returns to zero.

Explain these observations.

**(3 Marks)**

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**(b)** Explain how energy losses in a transformer are reduced by having:

**(i)** A soft-iron core.

**(2 Marks)**

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**(ii)** A laminated core

**(2 Marks)**

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**(c)** An ideal transformer has 2000 turns in the primary circuit and 200 turns in the secondary circuit. When primary circuit is connected to a 400V a.c. source the power delivered to a resistor in the secondary circuit is found to be 800W.

Determine the current in

(3 Marks)

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(ii) The primary circuit

(3 Marks)

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Figure 10 below shows a circuit in which a battery, a switch, a bulb, a resistor P, a variable resistor Q, a voltmeter V and ammeter  $A_1$  and  $A_2$  of negligible resistance are connected. P has resistance of  $10\Omega$ . When the switch is closed voltmeter reads 1.5V.

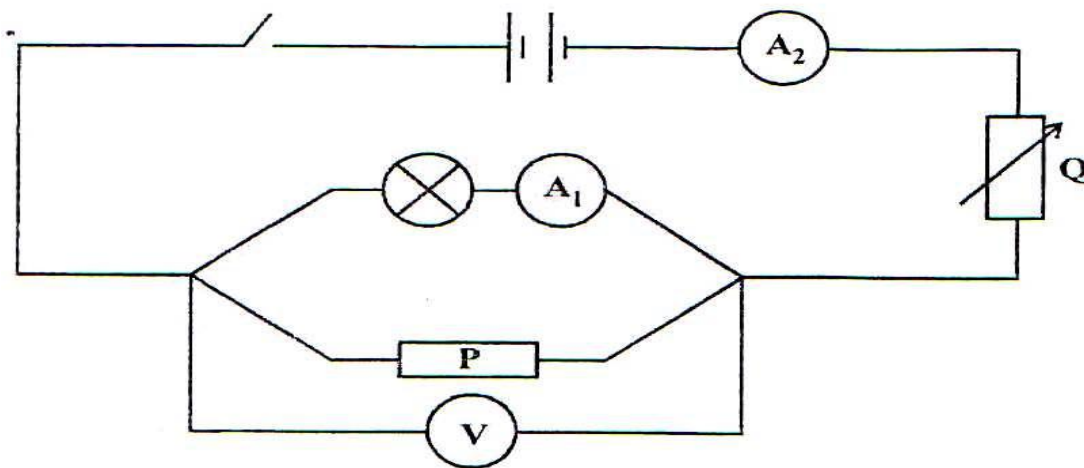


Fig10

(a) Determine

(i) The current passing through P, given that the current through it is 0.1 A . (3 Marks)

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(ii) the resistance of the bulb. (2 Marks)

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(b) The variable resistor Q is now adjusted so that a larger current flows through A<sub>2</sub>

(i) State how this will affect the brightness of the bulb. (1 Mark)

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(ii) Explain your answer in b (i) (2 Marks)

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